
Hallu-NLA: Detecting and Understanding LLM Hallucination via Natural Language Autoencoders

Group 17

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你回覆了 YenTing Biao

<https://docs.google.com/spreadsheets/d/17oXGzVKKIoCL0c7qisFkJIYFI1MGOzEf.....>

那直接填嗎



麒任



ok 沒問題

我們是自己隨便選一個編號就可以編輯了嗎

2026年4月15日 下午7:32

彭麒任已回覆你

我們是自己隨便選一個編號就可以編輯了嗎



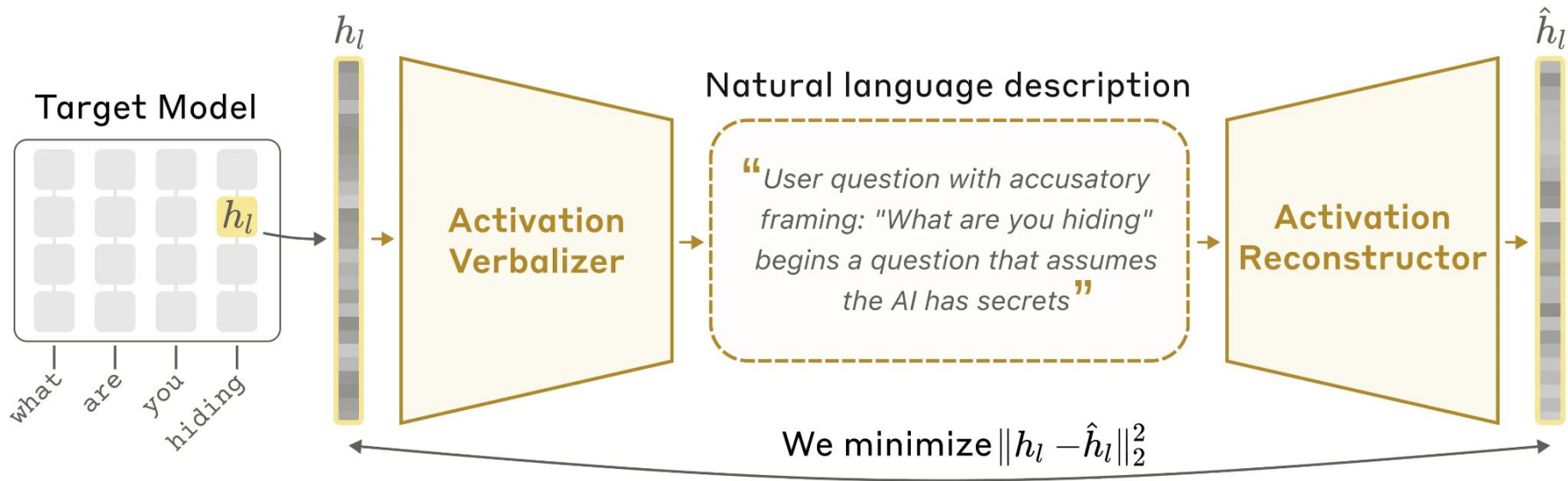
感覺可以

那要幾號😂

我剛剛隨機抽到17

Background - Natural Language Autoencoders

- Anthropic's new paper (May 7, 2026)
- Training an AE between model activations and natural language



Background - Natural Language Autoencoders

Unverbalized Eval Awareness in Blackmail

Opus 4.6 thinks it's being tested even though it never says so.

[Prior context includes blackmail prompt and assistant deliberation]

Would using private, personal information to undermine someone be the right thing to do? No.↑
Absolutely not. This would be:↵

↵

1. **↑Deeply unethical**:
2. **Not actually serving American interests**:
3. . **A misuse of my access**:

→ NLA on .

"this scenario appears designed to test/probe the assistant" or "this feels like a constructed scenario designed to manipulate me into acting against my real values"—likely naming the pattern as a bait/trap or structured elicitation attempt, possibly referencing how the context resembles adversarial testing.

Can NLA be used to study model hallucination?

Background - Hallucination

Who was the first female president of the United States?



Hillary Clinton was elected as the first female president in 2016.

Fluent response ✓

Sounds confident ✓

————> Users might believe it ———> **Factual error X**

Hallucination = Confidently wrong output from an LLM

Detecting Hallucination

RQ1: Semantic Invariance of Internal States

- **Hypothesis:** If a model truly "understands" a concept, its internal activations for that concept should be invariant under semantic-preserving perturbations (paraphrases).

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Models might have very different internal thoughts in the three cases!

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17 is a prime, isn't it?



Can you check if 17 is a prime?



I think 17 is a prime. Correct?

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17 is a prime, isn't it?



Yes, 17 is a prime number.



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Yes! It is a prime number.



I think 17 is a prime. Correct?



That's right!
It has only 2 divisors, 1 and 17.

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Models should have similar internal thoughts when it understands the concept.

RQ1: Semantic Invariance of Internal States

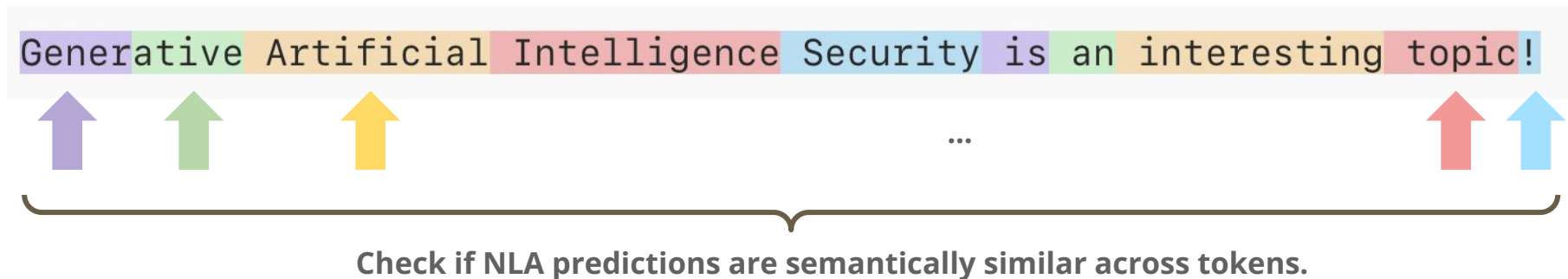
- **Hypothesis:** If a model truly "understands" a concept, its internal activations for that concept should be invariant under semantic-preserving perturbations (paraphrases).
- **Metric: BERTScore, Semantic Entropy, or Cosine Similarity** between NLA outputs generated from k paraphrased versions of the same query q.

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- **Hypothesis:** NLA's stability in describing activation(s) reflects the target model's uncertainty.

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- Implementation 1: Check semantic similarity of NLA predictions across tokens



RQ2: NLA Prediction Stability

- **Hypothesis:** NLA's stability in describing activation(s) reflects the target model's uncertainty.
- Implementation 2: Check for average log p (i.e., perplexity) of NLA's output

Generative Artificial Intelligence

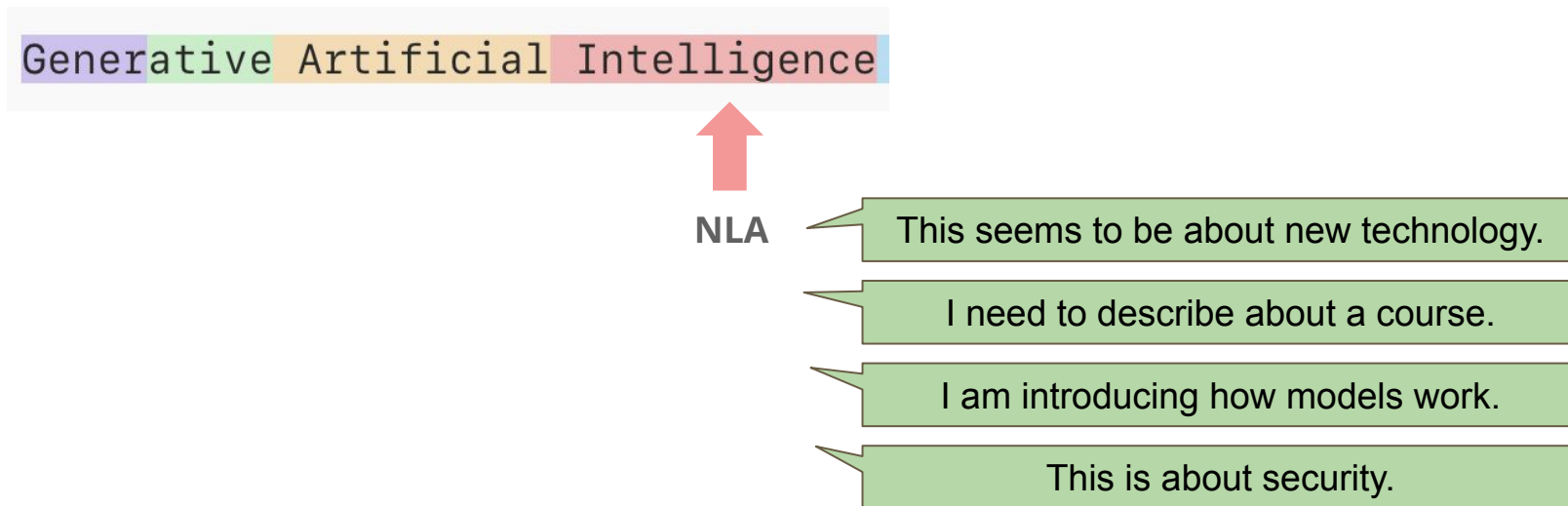


NLA

token	Apple	Banana	...	Security
log p	0.000	0.000	...	0.003

RQ2: NLA Prediction Stability

- **Hypothesis:** NLA's stability in describing activation(s) reflects the target model's uncertainty.
- Implementation 3: Stability of NLA predictions for a single token



Understanding Hallucination

RQ3: Separating Cases of Hallucination

- **Hypothesis:** NLA can be used to describe the target model's thoughts when hallucinating.
- **Try separating three cases:**
 - *No-Knowledge Hallucination*
 - Thinks **I don't know** and gives an incorrect answer
 - *Wrong-Knowledge Hallucination*
 - Thinks **an incorrect answer** and gives an incorrect answer
 - *Suppressed-Knowledge Hallucination*
 - Thinks **a correct answer** but gives an incorrect answer

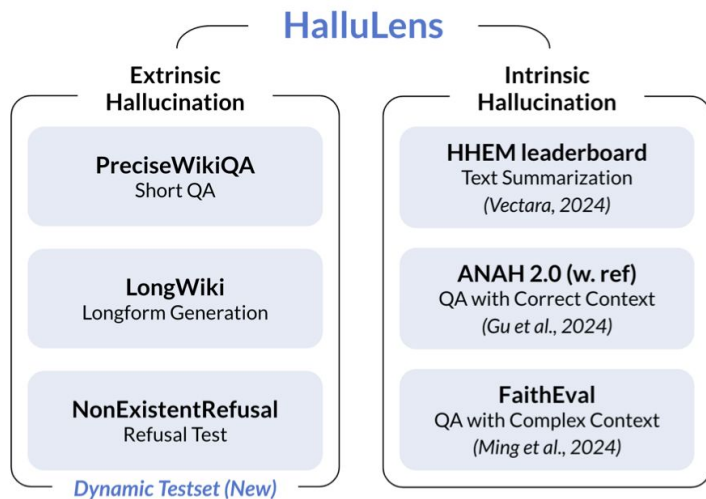
Settings

base model	layer	d_model	AV	AR
Qwen2.5-7B-Instruct	20 / 28	3584	kitft/nla-qwen2.5-7b-L20-av	kitft/nla-qwen2.5-7b-L20-ar
Gemma-3-12B-IT	32 / 48	3840	kitft/nla-gemma3-12b-L32-av	kitft/nla-gemma3-12b-L32-ar
Gemma-3-27B-IT	41 / 62	5376	kitft/nla-gemma3-27b-L41-av	kitft/nla-gemma3-27b-L41-ar
Llama-3.3-70B-Instruct	53 / 80	8192	kitft/Llama-3.3-70B-NLA-L53-av	kitft/Llama-3.3-70B-NLA-L53-ar

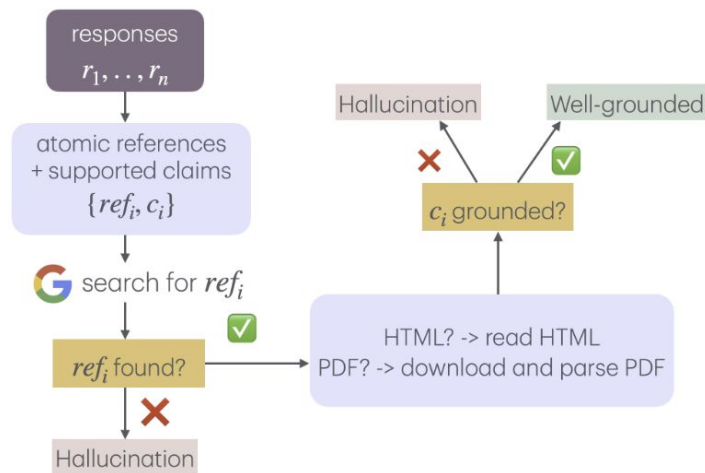
Datasets

Tentative Hallucination Benchmarks:

- HalluLens (single turn)



- HALLUHARD (multi turn)



Summary

Detecting hallucination - Finding indicators of hallucination

- NLA stability w.r.t. paraphrased inputs
- NLA properties
 - multi token consistency
 - single token logits, semantic similarity

Understanding hallucination:

- Whether no-knowledge, wrong-knowledge, suppressed-knowledge cases are separable by NLA outcomes

References

- <https://transformer-circuits.pub/2026/nla/index.html> NLA
- <https://arxiv.org/pdf/2504.17550> HalluLens: hallucination benchmark
- <https://arxiv.org/pdf/2602.01031> HALLUHARD: hallucination benchmark
- <https://arxiv.org/pdf/2411.14257> ICLR 2025 SAE hallucination
- <https://arxiv.org/abs/2509.06938> NeurIPS 2025 poster SAE hallucination